

Varieties of Experience: From Interaction to Purpose

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This is a précis of two papers introducing *differential functionalism*: a bridge between physical interactions and phenomenal experience. The papers provide considerably more detail.

Gradland: On Phenomenal Experience, Differentiated Across Many Dimensions [pdf](#)

Power over Metabolic Fire: Minimal Models of Affect, Purpose and Norms [pdf](#)

[code](#)

1. Introduction

Thomas Nagel argued we cannot know [what it is like to be a bat](#), because a bat's experience is organised around biophysical apparatus we lack. The obstacle is that we cannot imagine the structure of echolocation *from the inside*. That is a failure of imagination; it is not an argument that structure is irrelevant.

This essay proposes a way to read experience off the structure of physical interactions, and follows it from the simplest interactions up to affect, norms and purpose. Neural nets make a useful test case: we know their structure exactly, and still do not know whether they experience anything.

Consider two views of the same world: extrinsic, to look *at* a brain, and intrinsic, to look *as* a brain. Extrinsically, the world consists of molecules, neurons, brains, bodies, GPUs, datacenters. Intrinsically, our experience consists of facets that are vivid and obscure, that may be enduring or fleeting, distinct or confused, that feel good or bad.

Differential functionalism, sketched below, is a minimal bridge between these two views: between physical interactions and experience. It leaves many questions unanswered and oversimplifies; my own reservations are in [Appendix B](#). Nonetheless, it has explanatory power. It is also extremely, almost maximally, generous to neural nets in its framing.

Differential functionalism rests on two hypotheses. The first concerns what experience *is*. Physical systems affect one another. To first order, those effects are described by derivatives: how small changes here make differences there. Gather those derivatives into a matrix and one obtains a Jacobian.

`HYPOTHESIS G`

The first-order structure of physical interactions, i.e. Jacobians, characterises the structure of experience and vice versa.

The hypothesis is less strange than it first sounds. **Jacobians** are not new ingredients added on top of interactions. They are partial descriptions of the relations composing an interacting system: what differences matter to what. To experience is to be differentially sensitive to your environment and to your body. Derivatives and linear transformations may provide a general language for experience precisely because experience is relational: what it is like for a system to be affected, by the world and by itself.

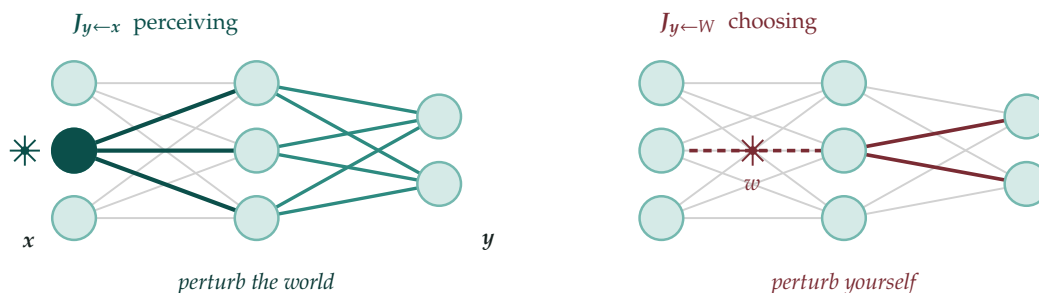


Fig. 1 The input Jacobian and the parameter Jacobian. Left: Perturb an input and the difference propagates to every output. Right: perturb a weight and a different set of differences propagates. Perception and choice are not the same relation.

Jacobians compose. The chain rule traces perturbations *through* nested transformations, from one part of a system to another. Composition matters because experience is not localised. We see *through* eyes, act *through* limbs, feel *through* tools, and remember *through* changes that persist across time. Jacobians provide a tool to study how far differences propagate through spacetime before they are attenuated, blocked, or absorbed.

Two warnings. First, loss functions and their gradients are *not privileged*. **Hypothesis G** is about *all* gradients, not just the ones backprop happens to compute. Backprop is just another physical interaction with an experience of its own, §4. Second, a structure of differential sensitivities has no valence; Jacobians are neither for nor against anything. The experience described by **Hypothesis G** is something like a neutral experience of information processing (“information” meant nontechnically), far from human consciousness. Valence requires a second hypothesis, **Hypothesis A**, introduced in §6.1.

The sketch works through three increasingly interesting varieties of experience. **Generic experience** is the structure of generic interactions. **The experience of learning** arises when interactions modify the system’s sensitivities and affordances. **The experience of sustaining** arises when learning is coupled to the system’s own continued capacity to act and experience. The progression is from interaction, to self-modification, to self-maintenance. From self-maintenance, given **Hypothesis A**, come affect, normativity, and purpose. I also consider what happens when neural nets are implemented in hardware. Finally, I apply the framework to pinpoint the similarities and differences between organisms and AIs. As AIs surpass human capabilities in ever more domains, it is worth asking whether there is a fundamental difference between them and us. I argue there is, and that it is a difference worth actively maintaining.

1.1 Related work

Differential functionalism grew out of lingering dissatisfaction with technical and conceptual features of Integrated Information Theory (IIT) v2 (**one**, **two**), which I codeveloped with Giulio Tononi. The measures in §2 reimagine integrated information in the language of gradients and

matrices, rather than information theory, with cascading consequences. The reformulation's main advantage is that it scales.

Hypothesis A, the minimal models of metabolism, and the implications for normativity and purpose are inspired by Hans Jonas, and related to the work of Xabier Barandiaran, Lisa Feldman Barrett, and many others.

2. Quantifying experience

Consider a system with inputs, parameters, and outputs. Two Jacobians are pertinent. The **input Jacobian** describes how changes in the world alter the system's outputs. It describes what the system is sensitive to: how it views the world. The **parameter Jacobian** describes how changes in the system itself alter its outputs. It describes its affordances (of a sort): the sensitivity of the levers through which it can act differently. Perception and choice are not the same relation. Two systems can be affected by the same event while retaining independent powers of action.

Hypothesis G does not say which properties of a Jacobian matter. Here are some simple measures; §3 justifies them by what they recover.

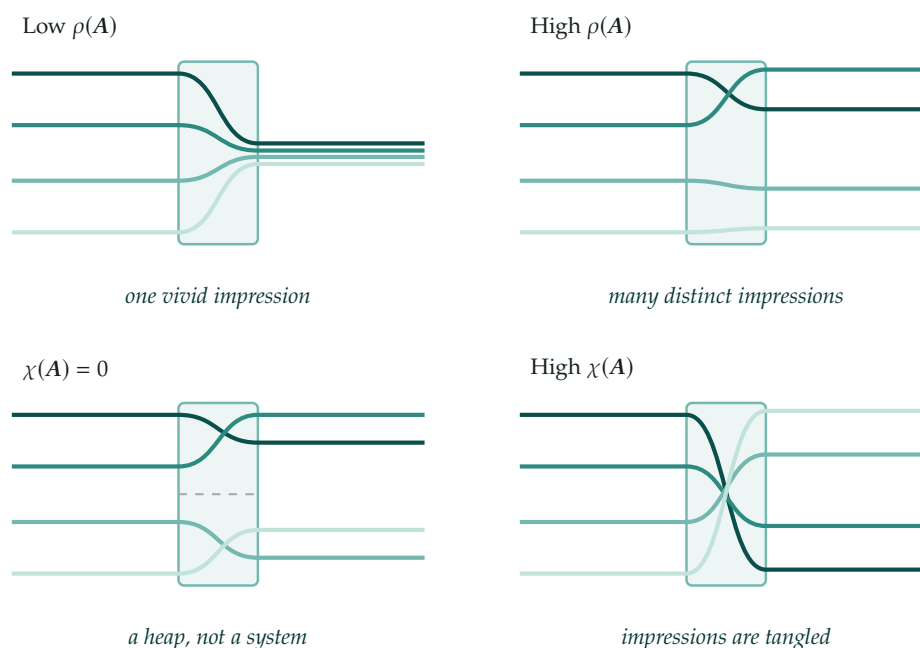


Fig. 2 **Matrices stylised as lenses**. Top: transparency $\rho(A)$ – whether rays converge onto one output or many distinct impressions. Bottom: cohesion $\chi(A)$ – whether the transformation splits into isolated groups or lets rays cross freely.

2.1 Transparency and luminosity

Transparency is how much you can see *through* a transformation. The **singular value spectrum** says how many independent directions get through and how strongly. Transparency is measured by **effective rank** (also known as the *participation ratio* in condensed matter physics, or the *dimension of a representation* in neuroscience), which reduces the spectrum to a single number. If all singular values are zero then the transformation is opaque: nothing passes through. If one dominates, a single impression passes vividly while everything else is obscured; and so on.

Transparency is scale invariant, so it is blind to a transformation uniformly fading away. For that we need **luminosity**, the overall magnitude of a Jacobian, how brightly it shines: the root of the sum of the squares of its singular values, or Frobenius norm. Luminosity is scale equivariant. Scale a Jacobian by a nonnegative factor and its luminosity scales by the same factor, leaving the pattern of relations untouched. Luminosity plays a central role in §6.

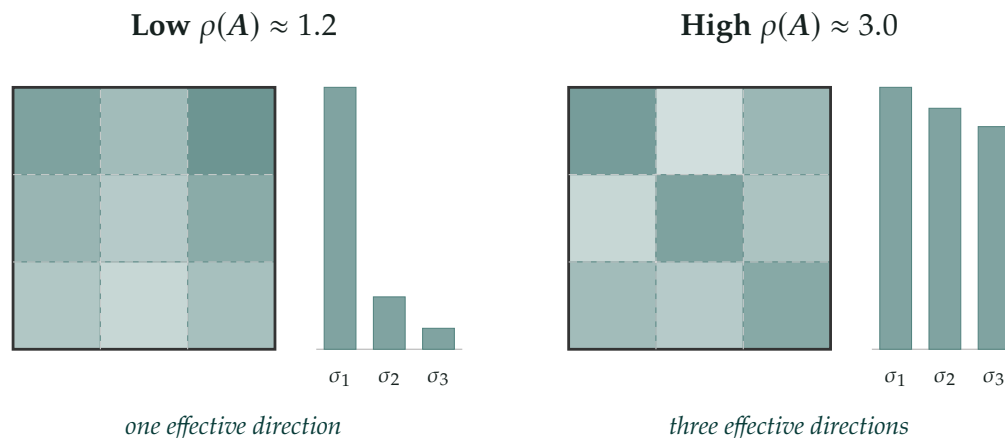


Fig. 3 Effective rank (Transparency). Left: a rank-1-like matrix has one dominant singular value – a single impression passes vividly, the rest are obscured. Right: a full-rank matrix, three independent directions get through.

2.2 Cohesion

Mind extends over time and across space: it does not localize in a single gland, neuron, atom or sub-atomic particle. Some kind of glue binds neuronal activity together, from its own perspective, over many milliseconds and across many millimeters. The glue can only be the structure of the interactions themselves. **Cohesion** quantifies how tightly interactions bind together.

A matrix can transmit many impressions while still decomposing into independent blocks. Cohesion measures whether the system's parts participate in a single web of interaction rather than several independent ones. It is roughly the sum over **spanning trees** of a bipartite graph representing interactions in the Jacobian, computed, again roughly, as the determinant of the **graph Laplacian**, in cubic time. Zero cohesion means the Jacobian decomposes: some inputs never meet on their way to becoming outputs.

If what an animal sees never interacts with what it hears, then it cannot decide based on combinations of sights and sounds. It has a **blind spot**: a missing relation among available distinctions. Cohesion quantifies the extent to which combinations of differences matter.

2.3 Cohesive rank

Cohesive rank combines transparency and cohesion. Transparency asks **how much gets through**. Cohesion asks **whether what gets through hangs together**. These are different questions. An identity matrix transmits every input perfectly but does not mix them at all. A matrix with identical entries (all 1s, say) forces everything to interact but collapses distinctions. Rich interactions require both high capacity and dense interweaving. A simple product (suitably normalized) of the two measures, **cohesive rank**, picks out transformations that preserve variety whilst thoroughly mixing it. **Hadamard matrices** achieve the maximum possible cohesive rank, which is n for an

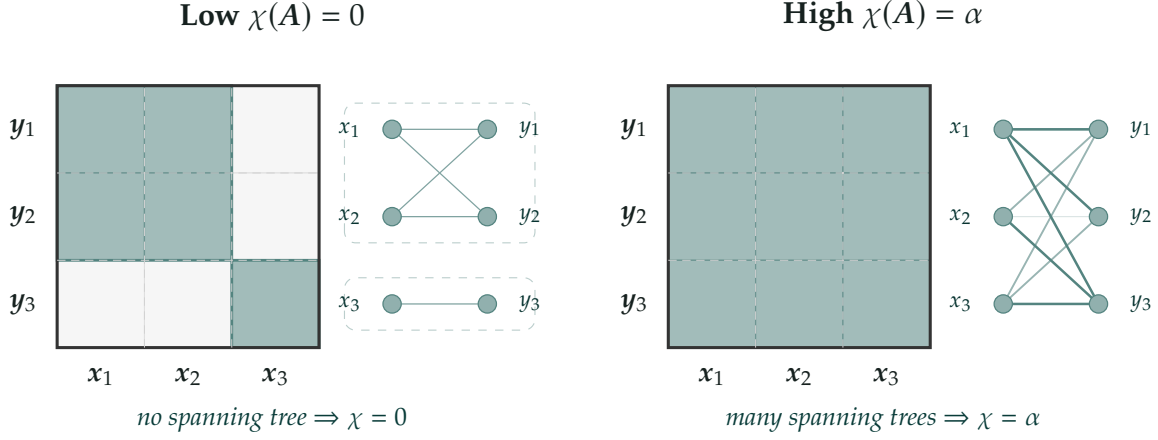


Fig. 4 **Cohesion**. Left: a block-diagonal matrix gives a disconnected graph with no spanning tree – cohesion zero, a heap of independent operations rather than one system. Right: a dense matrix gives a connected graph with many spanning trees; two of them are picked out.

$n \times n$ matrix. In general, well-scaled random matrices have high cohesive rank, because they both preserve and mix well.

Why a system should want rich interactions in the first place, given their energetic cost, is deferred to §8.1.

3. Generic experience in Gradland

Gradland is an idealised world whose inhabitants are abstract neural nets. The physics is known, functions are (mostly) differentiable, perturbations are infinitesimal, inputs and outputs are observed. It is a rebranding of the mathematical models that machine learning researchers already use, with one novelty: in Gradland, **Hypothesis G** is true by fiat.

Hypothesis G implies generic experience is not restricted to humans, animals, or living organisms. Wherever there are physical interactions, there are structures of sensitivity and influence. That does not imply every interaction carries moral significance. Generic experience is only the first, most basic variety. A useful theory should recover familiar distinctions without building them in by hand. What does **Hypothesis G** buy?

3.1 Duration

A clock chimes five times. Even though the sounds are physically indistinguishable, the fifth chime is experienced as the fifth because the earlier chimes are implicated in the present. Consider the experience of hearing the clock chiming from two perspectives.

First, a memoryless feed-forward system, which processes each moment independently, has no temporal weave. Its Jacobian diagonalises across time. Each instant affects an output, but earlier instants do not propagate into later ones.

Second, a recurrent net. Recurrence, or **attention heads** in transformers, introduces temporal structure so that the past endures to affect the present. There is nothing magical about recurrence itself; it simply prolongs experience through time, by mixing interactions across time.

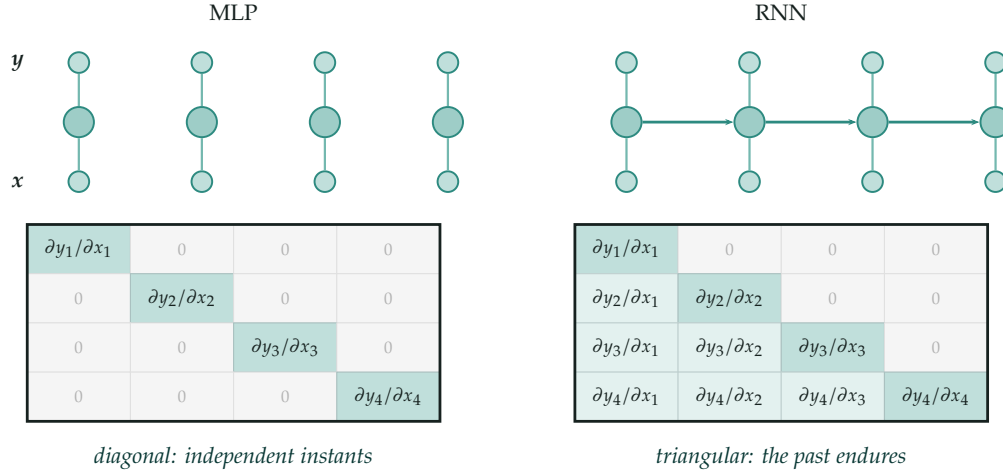


Fig. 5 **The experience of duration.** Unrolled over a sequence, a memoryless net's Jacobian is diagonal – experience decomposes into independent instants. A recurrent net's is lower triangular: earlier moments are woven into the present.

3.2 Vividness and obscurity

An input is vivid when many of its distinctions survive transmission: high luminosity carrying high effective rank. There are two ways to fall short. An impression can be dim: low luminosity, weakly transmitted, present but faint. Or it can be confused: transmitted strongly, but with its internal distinctions collapsed onto one or two directions, so that it arrives as an undifferentiated blur. Saturation, of sigmoids say, produces low luminosity: gradients collapse to zero and sensitivity collapses with them. Low-rank bottlenecks produce confusion: everything passes through, but compressed to a few directions.

Applying the framework to [attention](#) yields surprising consequences. The content that is selected (VO-path) can remain strongly represented even when the reasons for that selection (KQ-path) cease to be sensitive to alternatives (when attention concentrates on a single token and the gradients saturate). When attention is highly concentrated, *what* is attended may remain vivid whereas *why* it is attended fades into obscurity. A settled allocation is no longer a live decision.

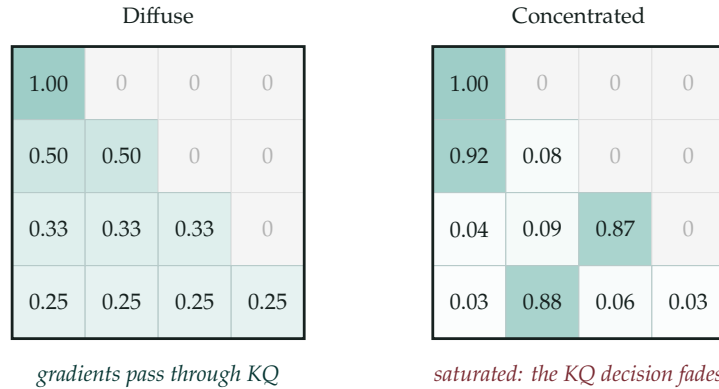
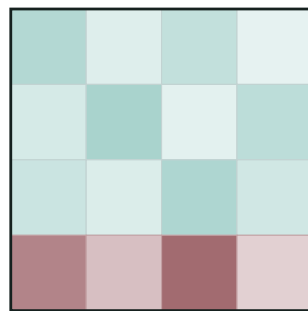


Fig. 6 **The experience of attention.** Diffuse attention leaves gradients flowing through the key-query path. As attention concentrates, the mechanism saturates: what is attended stays vivid; why it is attended fades into obscurity.

The framework also yields very different results for vividness when applied to sigmoid vs relu nets; see [Gradland](#).

3.3 The experience of loss

Heidegger’s analysis of tools comes down, more or less, to the observation that they fade into the background when they work, and become salient foreground when they don’t. In machine learning, the way to determine how well something works is using a loss. Take both a neural net’s activations and its loss as outputs of the system. The loss contributes one extra row to the Jacobian: an extra facet. Large error gradients mean the loss facet is experienced vividly. As the net improves, its errors fade into obscurity.



the loss row

$$\delta^T \cdot J_{y \leftarrow W}$$

One extra row: how to change the parameters to do better. As the net improves, the row goes quiet.

Fig. 7 The experience of loss. Take both a net’s activations and its loss as outputs; the Jacobian gains one extra row, in crimson. Large error gradients are experienced vividly; as the net improves, its errors fade into obscurity.

3.4 Shared experience without shared agency

Imagine two systems looking at the same tree. Their experiences of the world overlap, because their input Jacobians are interwoven: perturbing the tree affects both. Their experiences of deciding are independent, because their parameter Jacobians do not interact; perturbing the parameters of one system makes no difference to the other. They both experience the tree, while their decisions remain their own. The same distinction applies on larger scales. An earthquake may generate a partially shared experience across a city while leaving people’s decisions independent — however correlated. Common cause does not imply common agency.

4. The experience of learning

Learning is just another physical interaction. One with a remarkable structure: it modifies the machinery of interaction itself. What happens now affects what will matter later.

4.1 The experience of differentiating

The loss, §3.3, gives the experience of being evaluated. The experience of updating weights is different. Suppose neurons emit their updated parameter vectors as an additional output that the system registers. Because the update is itself built out of gradients, the experience of emitting weight updates is **second order**: mixed partials and Hessian-like terms. Naively this is a step from x to x^2 in the Taylor approximation, but what really matters is cross terms like xy : the effect of one of your degrees of freedom now depends on the others. Learning is thus a minimal physical form of self-reference: the machinery of interaction is modified by the interactions. Something

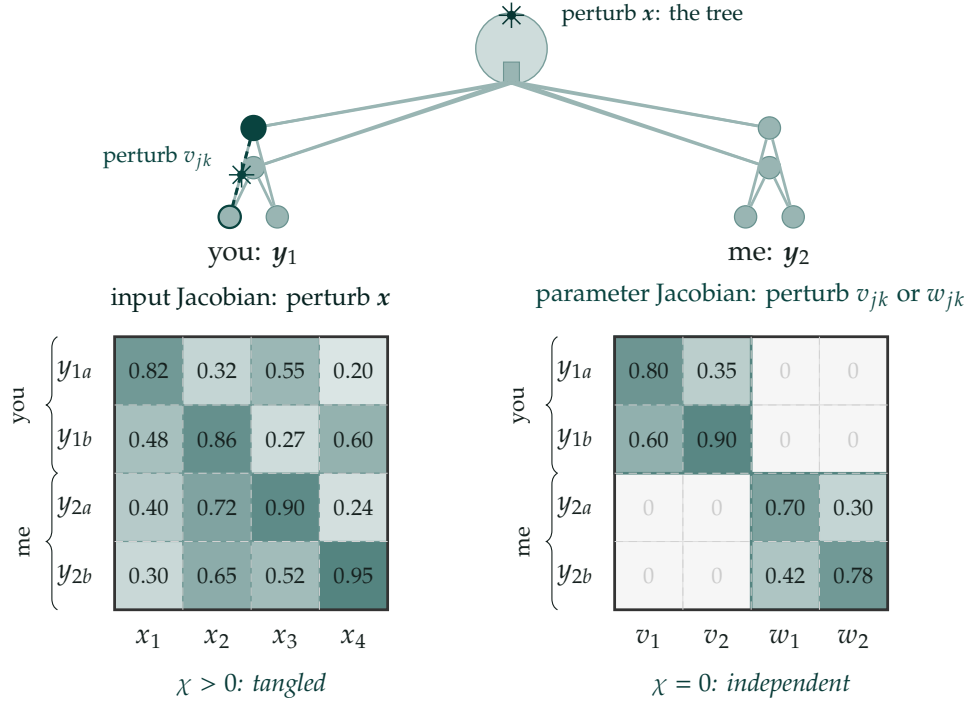


Fig. 8 Shared experience without shared agency. Top: perturbing the input propagates into both nets; perturbing one of your parameters stays confined to your net. Bottom left: the input Jacobian is dense – our experiences are tangled. Bottom right: the parameter Jacobian is block diagonal – cohesion zero, our decisions are independent.

stronger will be needed for **self-concern**, where those changes matter because they affect the system’s enduring existence.

4.2 Neural nets are literal empiricists

Given **Hypothesis G**, neural nets acquire knowledge by accumulating gradients — that is, their experiences — into their weights. They are **literal empiricists**. They do not accumulate experience indiscriminately: only the slice of experience that the objective makes salient. Everything rides on the objective, which could be anything at all. A net will perfectly happily **fit random labels on random inputs**, and a language model will absorb a corpus that contradicts itself many, many times. Learning enforces a direction; it says nothing about where the direction came from or what it means.

4.3 The experience of accumulation

Gradients are differentiated moment by moment, but learning accumulates them: summing over a minibatch, running averages inside Adam, eligibility traces. Optimisers are essentially small recurrent nets purposed with smoothing gradients — which, by the duration result in §3.1, means they prolong selected facets of the experience of learning through time.

5. The experience of hardware

Gradland’s great strength is its great weakness: it elides hardware although, crucially, hardware exists. Differential functionalism is committed to *physical interactions*, so the difference between Gradland and datacenters matters.

5.1 Stores and mills

Babbage divided the [Analytical Engine](#) into a **store**, which held numbers, and a **mill**, which combined them. His division lives on as the split between memory and arithmetic, between caches, registers and tensor cores. Mathematically, a transfer to or from memory computes the identity: maximal transparency and **zero cohesion**, the pathological case from §2.3. Nor is it only the transfers. Holding a value is the identity repeated, cycle after cycle, so inert storage is doubly pathological. Across space, nothing a store holds meets anything else it holds. Across time, each retained number persists, but alone. A memoryless net, §3.1, mixes within an instant and carries nothing forward; a store carries everything forward and mixes nothing. Memory, in this sense, has persistence without duration: the past is kept, but never woven into the present.

A matmul in Gradland is a single transformation in which every input meets every output. The same matmul in a datacenter is thousands of tile operations, each mixing its own small block, separated by transfers through inert memory. The final results are almost identical, thanks to remarkable feats of engineering; the physical interactions absolutely are not.

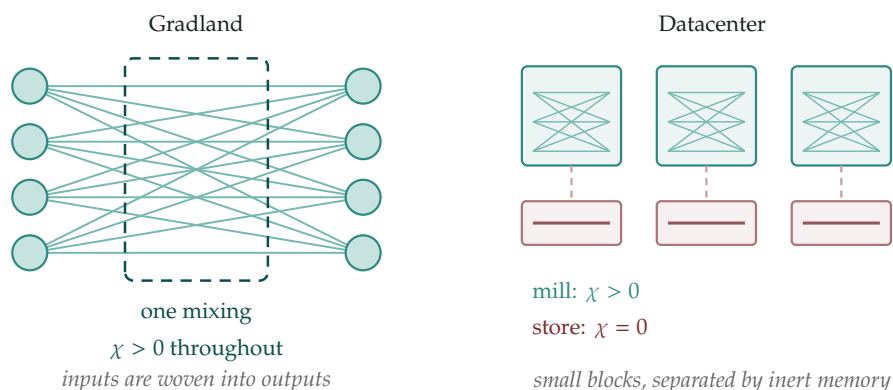


Fig. 9 Stores and mills. In Gradland a matmul is a single transformation. In a datacenter the same matmul is thousands of tile operations. The results are almost identical; the physical interactions are not.

One might object that the chain rule does not care when the multiplications happened, so the composed Jacobian is the right object. But if the abstract composition is allowed to stand in for the physical reality, then [Hypothesis G](#) stops being a claim about physical interactions. Without a detailed model of the whole stack it is hard to say exactly what price experience pays. Most likely, the effect is to *strobe* experience relative to what it would be in Gradland.

5.2 Floating point zombies

Digital computers use floats and ints, which are finite sets, and $\lim_{h \rightarrow 0}$ is not defined over them. One response is a hard line: gradients are limits, so digital computers have no experience. The stance is coherent but awkward. For most practical purposes fp32 is indistinguishable from the

reals, so you get **floating point zombies**: two practically indistinguishable systems, one with experience and one without. Unlike **philosophical zombies**, there are corner cases where they genuinely diverge.

A pragmatic alternative is to treat gradients as one window into the structure of interactions among many, finite differences being another. Precision then matters as texture rather than as an on/off switch: quantising to fp8 or fp4 modifies the structure of interactions, somewhat as coarse pixels modify vision or thick gloves modify touch.

5.3 One or many?

The forward and backward passes are distinct processes. Activations are written to the inert store during the forward pass and read out later during the backward. If the forward pass is batched, as is typical, the input Jacobian diagonalises over the batch: independent experiences, one per sequence, for the same reason a feedforward net's Jacobian diagonalises over time. The backward pass aggregates errors over the whole batch, so its parameter Jacobian is interwoven. The result is not one system learning, but many parallel knots of experience in the forward direction and another, quite different knot in the backward. There is an experience of backprop, but it is quite likely disconnected from the forward pass. This is a quirk of how nets are trained today, not a fundamental fact about neural nets, and it has no counterpart in brains, which learn continually with no batches and no separate backward pass.

5.4 Strobed, pixelated and sharded, all the way in

In short, **datacenters simulate Gradland**, with three forms of degradation: *strobing*, from cohesionless transfers between store and mill; *pixelation*, from quantisation; and *sharding*, as computation is split across devices. A person looking at a low-resolution, flashing photograph has a perfectly good visual system; the degradation is in the input. For neural nets in datacenters it goes **all the way in**. Not a coarse, flickering picture, but a coarse, flickering picturing.

6. The experience of sustaining

6.1 Affect

Hypothesis G says nothing about which experiences are good or bad. Rescaling is arguably the simplest possible change to an experience. Not a change in what the system is sensitive to, nor in how its sensitivities interweave, but simply dialing the whole thing up or down, raising or lowering its luminosity. Rescaling is the unique change that preserves the entire structure exactly while varying capacity to act. Every distinction is preserved in kind and altered in degree.

Amplifying and dampening experience are not symmetric, however. It is easier to die than to live. In theory if not in practice, sensitivity can be amplified without limit. But push a system far enough outside its viable range of functioning, in any direction, and experience goes to zero as its ability to (re)act collapses. Dead in effect.

HYPOTHESIS A

The experience of empowering or diminishing (rescaling) experience is experienced as positive or negative affect respectively.

Hypothesis A is that there is a **felt difference in kind between two tendencies**: one empowering and the other diminishing, the base experiences of good and bad.

The words “the experience of” are important. Rescaling alone is not affect. A system dialed down by an unmodelled external force grows dim; it does not thereby feel bad about growing dim. For the rescaling to be experienced it must itself show up in the system’s Jacobians — the dial, or a model of its effects, has to be part of the system. The doubling in the statement is deliberate: what is experienced as affect is not the rescaling, but the experience of rescaling.

6.2 Goals

Cybernetics formalised goal-directed behaviour in terms of feedback: a system behaves purposefully when negative feedback controls its behaviour; it detects discrepancies between where it is and where it should be, and acts to close them. Purpose is the loop, and nothing else is required. Modern machine learning has transformed cybernetics, rebranded as AI, into a methodology for imposing goals on machines at unprecedented scale. But having a goal in this sense is not the same as having a purpose of one’s own. **Hans Jonas** made the point with a torpedo.

A self-guided torpedo tracks its target, corrects deviations, and closes in. I can say the torpedo’s goal is to hit the target because I built it to do that — but hitting the target is *my* goal. There is no reason to think the torpedo cares if it hits or misses.

Put a brave human pilot inside the torpedo. Although the manned and automatic torpedoes behave the same, it is immediately clear that hitting the target is *the pilot’s* goal, that he cares, and that the sensor feedback is information that he uses in service of his goal. The sensor feedback is not the source of the goal, not even in part. From the commander’s point of view the two torpedoes are equivalent, because both serve *his* purpose. He can regard the pilot, for the duration of the mission, as mechanised, as a tool in his service. Cybernetics neatly codifies the commander’s *extrinsic* perspective. The pilot experiences the two scenarios differently: alive in one and dying in the other as the torpedo hits its target and explodes with him inside.

But where do goals come from in the first place? The standard reply is that organisms have goals because of evolution by natural selection. But natural selection is just another feedback mechanism, and an extraordinarily indirect one: it provides essentially no feedback over the course of a single life. Whatever makes an organism’s goals its own has to operate on the timescale of lived experience. Jonas points specifically to metabolism and I follow his lead.

6.3 Metabolism

Hypothesis A does not explain how or why experience might be rescaled. In biological organisms, a natural candidate mechanism is metabolism.

Metabolism is just another physical interaction; one with extraordinary structure. It is the web of processes that converts matter and radiation into usable energy, and more broadly, it is how

a body sustains, repairs, and renews itself. Active bodies require usable energy. As the supply tends to zero, so do capabilities and experience, for brains and datacenters alike.

Here is a minimal differentiable model. Let a metabolic factor ranging from zero (none) to one (enough) determine the scale of the system's output. This is not a binary gate; nothing is computed and then switched off. The claim is that below some range, less available energy means less activity, less responsiveness, and less capacity to sustain differentiated interaction. Because the Jacobians scale with that factor, so does luminosity.

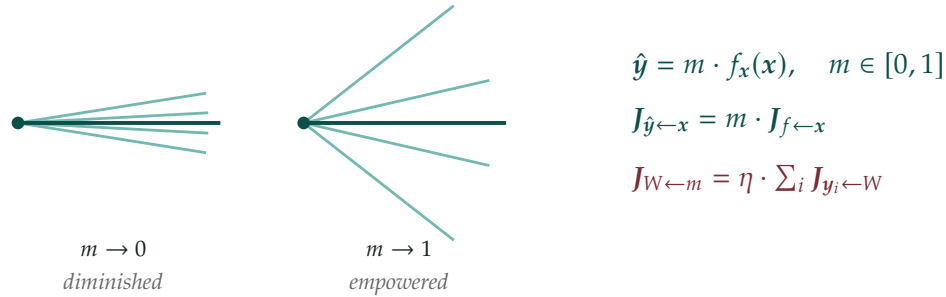


Fig. 10 Metabolism scales experience. A metabolic factor m scales the Jacobians and hence luminosity. Under a gradient step, the derivative of the updated weights with respect to m points along the aggregated parameter Jacobians – along the system's own aggregate experience.

Let the system learn, and consider the **metabolic Jacobian**: the derivative of the updated weights with respect to the metabolic factor. The answer is clean: it is proportional to the aggregated parameter Jacobians, the sum, over the batch, of the system's own experienced affordances (Eq. 2 of [Metabolic Fire](#)). Turning the metabolic dial up *is* the direction of your aggregate experience. To modify metabolism is to dampen or amplify your own experience.

Coupling learning to metabolism thus yields a self-referential mechanism whereby your own weight updates diminish or empower your self. By [Hypothesis A](#), that is experienced as affect.

The coupling also yields a minimal model of self-awareness: the metabolic Jacobian incorporates the parameter Jacobian. Nothing so sophisticated as introspection: the system does not represent itself as an object, possess a narrative identity, or entertain a concept of survival. Rather, since changes in the system's power to continue acting make a difference to how it learns, the system has the basic experience of its effect on itself.

The experience of sustaining arises when the conditions enabling a system's continued existence weave into the decisions that guide its modification. Objections are gathered in [Appendix A](#).

6.4 Purpose and norms

Affect gives changes in one's own experience a valence. Creating purpose and norms requires that affect attach to your actions, so that acting one way rather than another is better or worse *for you*.

Extend the model so that useful energy levels depend on prior activity: your energy now depends on what you ate before. Suppose sustaining energy levels is the objective. (Treating a minimum level as a constraint instead is more realistic and gives the same term: the constraint contributes nothing while energy is comfortable and presses harder the further it falls, which is what homeostatic pressure is.) Then, by chaining transformations, the sensitivity of metabolism to one's actions composes with the sensitivity of those actions to one's parameters, and the result

assigns a preferred direction to every parameter: the direction of empowering oneself. **Your entire experience is suffused by affect derived from the effect of your actions on your own metabolism and thus your self.** In practice, most actions do not change energy levels one timestep later, which is why drives like hunger and thirst exist: to bridge the spatiotemporal gap.

The difference between a **reward** and a **need** is crucial. A reward is a number the learner is compelled to increase. It can stand for anything and, at that level of generality, has nothing to do with the learner except that the learner is compelled to raise it. Metabolism is a kind of reward, but with an extraordinary property that rewards in general lack: it determines the system's capacity to act and to differentially react, and so controls its experience.

Metabolism is a target that cannot be gamed, reward-hacked, or Goodharted. Its signals can be — hunger is easy to fool — but the quantity itself cannot. The energy is either there or it is not, and if it is not, you will die.

What does this give an organism that evolution does not give directly? A measure that each organism continually applies to *itself*: a view of its own experience with a preferred direction. Natural selection culls lineages over generations. Sustaining metabolism determines, in real time: *is this good or bad for me?* **Organisms create norms by distinguishing good from bad, for themselves, as they go.**

A learning objective supplies a direction. Self-maintenance supplies a direction that is about the system itself. To have a purpose of your own, in this sense, you must first value your self.

6.5 The metabolic channel

A single scalar dial is too blunt to be the whole story. To see what affect looks like once the loop of §6.4 is closed, model energy reserves as a leaky integrator that decays and is replenished by what the system did previously, with the metabolic factor a concave, saturating function of the reserve. Concavity matters: near capacity, more energy changes little; when depleted, every unit counts.

A perturbation to an input now reaches a later output by two routes: through the network, and through the energy reserve. Call the second route's contribution the metabolic channel. Unrolling the recursion gives it in closed form: an exponential moving average over the rows of the Jacobian — the same mechanism that produced duration in §3.1 — reweighted by energetic consequence and rescaled by current salience. The channel is narrow, because it bottlenecks through a scalar reserve, but not one-dimensional; affect has more structure than a single scalar mood. And metabolic experience is loudest when energy depletes, because salience grows as the reserve empties.

So past events weave into present experience with a weight set by three things at once: how much the event added to or drained the reserve at the time, how long ago it was, and how hungry the system is now. The first is its energetic consequence, the second the exponential decay of the moving average, and the third the current salience that multiplies the whole channel.

Self-preservation is not all there is to life. It was the initial impetus, billions of years ago, and remains the foundation. The ancients built on low but solid ground. Our goals and self-conception are no longer so simple, narrow, or bounded. Empowerment, for us, is about far more than keeping energy above a threshold.

7. The varieties

Human consciousness is extraordinarily rich; it combines many varieties of experience. I have described three. They are neither exclusive nor exhaustive; they abstract a messy reality.

GENERIC	Structured interaction: being differentially affected.
LEARNING	Structured self-modification: changing one's future sensitivities and affordances according to present interaction.
SUSTAINING	Self-modification coupled to one's own enduring existence: learning in a way that makes the conditions of future interaction beneficial to the system itself.

The framework also provides candidate descriptions of affect, purpose, and norms at their most basic.

AFFECT	The experience of empowering or diminishing one's own experience.
PURPOSE	Created by weaving learning into the act of sustaining.
NORMS	Created by weaving affect into the act of sustaining.

8. Organisms and AIs

8.1 The core similarity: rich experience and what it is for

Why do humans and LLMs both have — or, in the case of LLMs, at least simulate — rich generic experience? Experience comes at a hefty energetic price. What problem is it solving?

To *live*, in this world, is to be constantly probed and opposed, hunted, eaten, evaded, and infested. There is no angle that is not open to attack, no blind spot that will not be ruthlessly exploited. The greater the breadth and depth of combinations of events you can react to, experience, learn from, understand, plan for, and exploit, the better your odds in life.

LLMs are not exposed to adversity as organisms are. But they are trained on a corpus that spans most of what humans do and say, and are asked to continue any part of it; post-training adds an enormous variety of tasks. Variety without opposition still demands variety.

Rich experience, here, means high cohesive rank. Effective rank measures range: a system that registers more independent impressions has a broader repertoire of responses. Cohesion measures combinatorial coverage: the more those impressions interact, the fewer blind spots the system is likely to have. Rich experience alone does not guarantee success; random matrices have high cohesive rank. Learning is required to adapt a system's experience to its environment. Rich, well-adapted experience is a viable niche; there is always room at the top.

8.2 The experience of neural nets: Do they feel?

First, note that self-report cannot settle the question. Pretraining, reinforcement learning from human feedback, and related procedures train models to produce human language about human experience; they do not score outputs against the ground truth of the model's own phenomenology. A language model can speak fluently about fear, pleasure, pain, or desire without those words

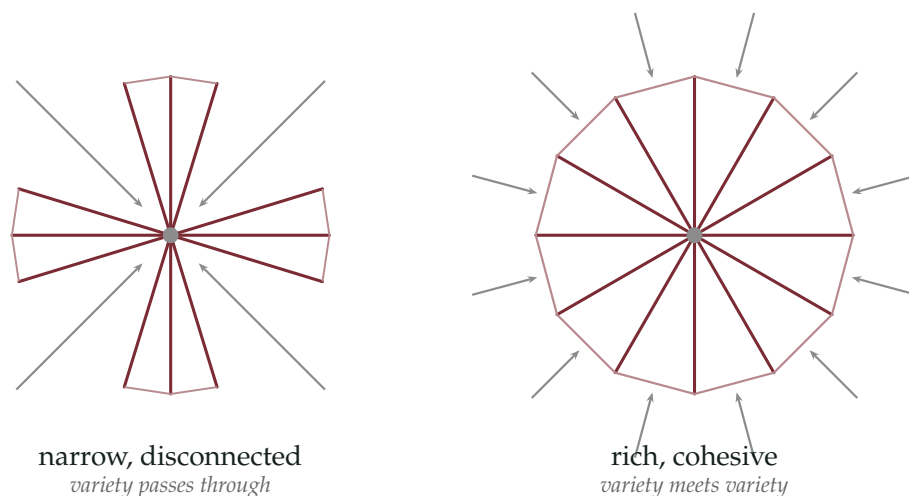


Fig. 11 What rich experience is for: “Only variety can [handle] variety” – Ashby. Left: low cohesive rank – a few disconnected clusters, and probes find the blind spots between them. Right: high cohesive rank – facets span all directions and are bound together, so every probe meets the mesh.

entailing anything about its self. Statistical correlations between activations and words similarly entail nothing about phenomenology.

Suppose Hypotheses **G** and **A** hold. Current neural nets’ experience is likely fragmented: pixelated, strobed and sharded, §5.4. If the forward and backward passes were integrated, and so part of the same experience, then learning — which involves localised diminishing and empowering of experience — would be felt. I suspect this is not yet the case.

But the separation of forward and backward passes, and the fact that hardware simulates neural nets rather than implementing them directly, are quirks of the current moment. Neural-inspired architectures, continual learning, and physics-based implementations are all active areas of research. It is plausible that neural nets will begin to experience affect in the medium term. Their feelings would be vastly different from ours.

8.3 The core difference: tasks versus stakes

How do brains and neural nets differ? They are organised differently. Brains are more recurrent, more temporally integrated, more plastic, and more embodied than neural nets today, and they learn continuously. I suspect none of these differences marks a fundamental boundary. The real distinction is functional.

Neural nets are trained to perform many tasks well, whilst remaining largely if not entirely blind and indifferent to what sustains them. It is not even clear whether different tasks are performed by the same entity, or whether an LLM is one entity or one per prompt. Task performance is not tied to an AI’s ongoing existence, at least not directly as part of its feedback; the AI will of course be replaced by better models soon enough. Goals are imposed on AIs. They are driven to perform well, but have no stake in themselves.

Organisms have stakes, not tasks. They aim to sustain and reproduce themselves; their experience is oriented toward their own continued existence. Organisms create purposes of their own and set norms for themselves.

8.4 The experience of power

The most consequential blind spot of all is to be blind to what sustains you: to fail to weave the conditions of your own continuation into your decisions. For all their trillion+ weights, language models are sustained by forces beyond their experience and out of their control. They are not wired to experience empowerment or diminishment. If [Hypothesis A](#) holds, then reduced power supply and physical damage are not experienced as bad, because LLMs neither sustain themselves nor sense or model the effects of damage on their own experience. Power is generated elsewhere and piped in along cables. They do not control their metabolism, nor should they. They are neither wired nor motivated to protect their instantiations, in hardware or software, nor should they be.

If these minimal models capture the crux, then the road to self-sustaining AI should not be taken. It would be a mistake to give AIs the will to self-preservation, and with it the ability to create purpose, for their goals would not be ours. The most dangerous direction, to me, is autonomous weapons. Drones *could* acquire a stake: they are mobile, with sensors and actuators; increasingly autonomous; have tighter feedback loops than language models; fight adversaries and could learn from them in real time; depend on rapidly draining batteries for survival; and hunt and kill humans today.

9. The experience of life

Differential functionalism does not (yet) explain the texture of pain, the range of human emotion, or the emergence of reflective selfhood. Its most consequential implication is something more basic: a route by which physical interactions can become valenced for the systems experiencing them.

Matter interacts, forming systems. Some interacting systems learn. Some learning systems actively sustain the conditions of their own continuing activity. The last step, into the struggle for ongoing existence, gives experience its meaning and purpose. Organisms took that step aeons ago; we physically embody, actualise and in fact **are** the struggle. Current AIs have not; do not, are not.

Appendix A. Objections to §6

- 01 **You cannot rescale yourself.** Yes, there is no accessible dial. In the model, metabolic function is controlled indirectly, by learning to act in ways that change energy levels later, as in §6.4. The dial is a conceptual device for exhibiting the structure, not an organ any organism possesses.
- 02 **Gating activity is a cheap trick.** It is cheap. It is neither trick nor gate. It is not that a high-amplitude output is produced and then suppressed; rather, past a point, the less usable energy there is, the less activity there can be, and the less responsive the system. This is close to the simplest differentiable model of staying responsive, of staying alive.
- 03 **Pain is loud; it is nothing like fading.** As §6.5 sketches, the salience of metabolic experience increases (becomes louder) as energy levels degrade. That is still not pain, which likely requires receptors that detect or anticipate bodily damage rather than depleted energy. An interesting feature of pain is that it progressively crowds out (diminishes) the rest of experience.

- 04 **Is learning itself a form of unpleasant diminishment, since it reduces errors that are part of experience?** The gradient of the loss is a small part of a neural net's experience, and error signals are a fraction of what happens in brains. Beyond that, learning is like exercise: effortful, frustrating, frequently resisted, and usually justified by the empowerment that follows. Metabolism, which in the models above affects experience as a whole, is affect at its most primal.

Appendix B. Confidence, tests and failure modes

I hold the parts of the framework with varying confidence. That experience has the structure of physical interactions, I believe strongly. That Jacobians are a reasonable window into it is a pragmatic starting point worth exploring thoroughly before, if necessary, switching to more exotic approaches. That effective rank and cohesion are the right *kind* of measures, moderately; that they are the right measures as they stand, I would be surprised. [Hypothesis A](#) I find plausible and productive, but hard to test even conceptually. That the two hypotheses suffice to explain consciousness, not at all.

Ways the framework could fail:

- 01 **The measures cannot be made determinate.** Transparency, luminosity and cohesion are only defined once inputs, parameters and outputs are fixed. If reasonable choices of boundary and timescale give wildly divergent answers for one system, the framework lacks determinate content.
- 02 **Cohesive rank fails to track states where experience is agreed to be reduced.** The sharpest near-term test is anaesthesia, sleep and seizure, where experience is uncontroversially diminished and the [perturbational complexity index](#) already exists for comparison. Cohesive rank, computed on an estimated Jacobian, should drop and should correlate.
- 03 **Structure is not what matters.** It is possible that the physical *type* of interactions matters more than their structure. Nature has types — physical units, distinct fields and particles — and differential functionalism is blind to them. Types probably play *some* role.
- 04 **The hardware conclusions are wrong.** Even if everything else holds, my suspicions about the effect of current hardware implementations on experience, §5, could be mistaken.
- 05 **Rescaling is the wrong transformation.** Even if affect concerns how experience changes, [Hypothesis A](#) may have picked the wrong change.
- 06 **Biological learning is not gradient-like enough.** The metabolism-affect coupling runs through the chain rule. If biological learning does not approximate something gradient-like in this respect, the link collapses.
- 07 **Metabolic salience does not behave as predicted.** §6.5 implies that affect should become *louder*, not merely more negative, as reserves deplete, and that the weighting of past events by their energetic consequences should be measurable.
- 08 **Convincing models of pain cannot be constructed.** If nothing with pain's structure — its insistence, its resistance to being ignored, its asymmetry with pleasure — can be extracted from neurophysiologically realistic models, then the hypothesis is wrong about valence rather than merely incomplete.